TCW 1103 · ENGINEERING COMMUNICATION SKILLS · PART 1

# Communication Theory and Interpersonal Skills

The process, the models, the channels, the barriers, listening, non-verbal communication, assertiveness, emotional intelligence, conflict and leadership. Written for the exam: summaries, spotted questions and model paragraphs.

**WHITEPAPER 1 OF 3**

COURSE
**TCW 1103 Engineering Communication Skills (Part 1)**

DEPARTMENT
**Civil & Water Engineering, NUST Faculty of Engineering**

LECTURER
**Mr T. L. Luzane**

PREPARED FOR
**Group 13 · exam on Monday 26 October 2026**

INSIDE THIS WHITEPAPER

1. Communication and engineering communication defined
2. Elements and process (with the diagram to draw)
3. Models: Shannon-Weaver, Schramm, Berlo, transactional
4. Verbal and non-verbal communication; appearance
5. Channels and directions in an organisation; public relations
6. Barriers and how to overcome them
7. The Seven C's
8. Listening versus hearing; active listening
9. Assertiveness, manipulators and put-down offenders
10. Emotional intelligence and self-regulation
11. Conflict resolution and Kurt Lewin
12. Leadership theories; transformational leadership
13. Exam spotter and practice questions

Sources: lecturer's TCW 1103 topic list and class presentations, the NUST TIE 1102 papers (Dec 2014, Aug 2015), the group whitepaper, and the standard textbooks listed at the end.

Group 13 Study Board

**HOW TO USE THESE THREE WHITEPAPERS**

Each whitepaper is short on purpose. Every section ends with an **Exam spot** box (what the examiner wants) and most have a **model paragraph** you can adapt. Stars show how likely a topic is: HOT very likely, LIKELY regular, POSSIBLE could appear. Whitepaper 3 ends with the full exam spotter, a 40-question bank in past-paper style and the Monday plan.

# 1 Communication and engineering communication

**Communication** is the process of transmitting information, ideas, attitudes or feelings from one person or group (the sender) to another (the receiver) so that the receiver understands the message as the sender intended and responds. The word comes from the Latin *communicare*, "to share" or "to make common". The key test is **shared understanding**: if the receiver did not understand, communication has not taken place, only transmission.

**Engineering (technical) communication** is the transfer of specialised technical information, such as designs, calculations, specifications, procedures, test results and safety requirements, between engineers, technicians, clients, contractors, regulators and the public, in a form that is accurate, clear and usable. It differs from everyday communication because it is **technical** (abstract ideas must be translated for non-specialists), **consequential** (an ambiguous drawing note or safety instruction can injure people or waste money) and **multi-channel** (drawings, reports, memos, meetings, site instructions, email, BIM models).

### Purposes of communication

- To **inform** (progress report, notice)
- To **instruct** or direct (site instruction, SOP)
- To **persuade** (proposal, tender, presentation)
- To **request** (memo asking for resources)
- To **build relationships** and morale
- To **record** (minutes, drawings, contracts)

### Why it matters to an engineer

- **Safety**: clear instructions prevent accidents
- **Efficiency**: fewer clarifications, less rework
- **Teamwork**: architects, QS, contractors, clients
- **Credibility**: clear reports win trust and promotions
- **Decision-making**: management acts on your reports
- **Legal record**: drawings, minutes and letters are evidence

**EXAM SPOT** HOT

"Define (engineering) communication and explain its importance" opens many papers (NUST TIE 1102 Aug 2015 Q1; Dec 2014 Q1). Give a **definition with the shared-understanding test**, then 4 to 6 points of importance, **each with an engineering example**. One sentence per point is not enough at 10 marks: explain and illustrate.

# 2 Elements and the communication process

Communication is a **cycle**, not a single event. A sender has an idea, encodes it, sends it through a channel, the receiver decodes it and sends feedback. Noise can interfere at any stage.

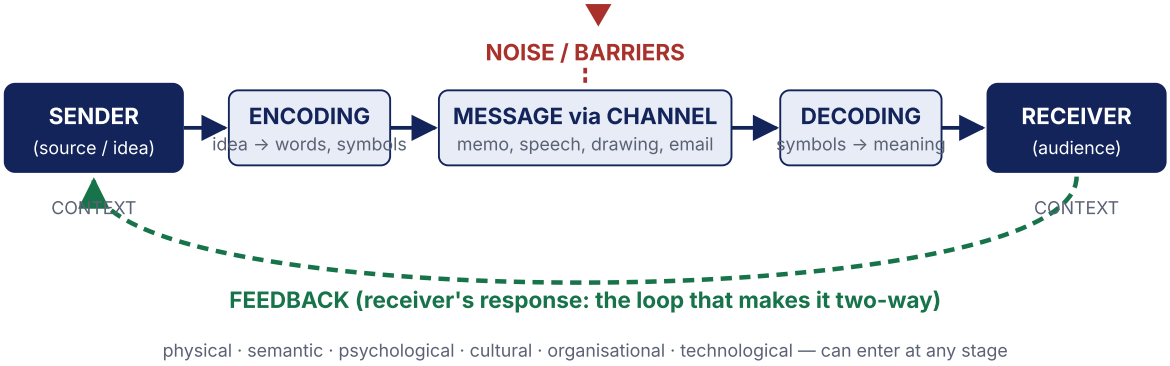



Figure 1. The communication process. Draw this in the exam with every label: examiners mark the diagram and the explanation separately.

| Element | Meaning | Engineering example |
|---|---|---|
| **Sender** | Originator of the message; their knowledge, attitude and skill shape its quality. | Site engineer with a change to the rebar schedule |
| **Encoding** | Turning the idea into words, symbols, numbers, gestures or a drawing. | Revising the drawing and writing a site instruction |
| **Message** | The content: the information, instruction or request. | "Use Y16 bars at 150 mm centres on beam B3" |
| **Channel / medium** | The route: face to face, phone, memo, email, drawing, meeting, notice board. | Revised drawing plus toolbox talk |
| **Receiver** | The person or group the message is for. | Site foreman and steel fixers |
| **Decoding** | The receiver interpreting symbols back into meaning using their own knowledge. | Foreman reads the drawing and bar schedule |
| **Feedback** | Receiver's response, showing whether the message was understood. Makes communication two-way. | Foreman repeats the spacing back; asks about cover |
| **Noise** | Anything that distorts the message: physical, semantic, psychological. | Concrete mixer running during the briefing |
| **Context** | The setting: physical, social, cultural, time. | Urgent pour scheduled for tomorrow morning |

# 3 Models of communication

| Model | Type | Key idea | Strength / weakness |
|---|---|---|---|
| **Aristotle** (c. 350 BC) | Linear | Speaker, speech, occasion, audience, effect. Persuasion by *ethos* (credibility), *pathos* (emotion), *logos* (logic). | Good for public speaking; ignores feedback and noise. |
| **Shannon and Weaver** (1949) | Linear | Information source → transmitter → channel (+ noise source) → receiver → destination. Built for telephone engineering. | Introduced **noise**; one-way, no feedback. |
| **Berlo SMCR** (1960) | Linear | Source, Message, Channel, Receiver. Success depends on the communication skills, attitudes, knowledge, social system and culture of source and receiver; channel = the five senses. | Explains why the same message lands differently; no feedback. |
| **Osgood and Schramm** (1954) | Interactive / circular | Both parties encode, decode and interpret. Communication works where their **fields of experience overlap**. | Adds feedback; assumes equal roles. |

| Model | Type | Key idea | Strength / weakness |
|---|---|---|---|
| **Transactional** (Barnlund, 1970) | Transactional | Both parties send and receive **at the same time** (words plus non-verbal), within a shared context. | Most realistic for meetings and site talks. |

**EXAM SPOT** HOT

"With the aid of a diagram explain the elements of communication" [12 marks, NUST Dec 2014]. Draw Figure 1, then explain each element in order with an example, and end with: *feedback turns transmission into communication*; *noise can enter at any stage*. If asked for a **model**, name it, date it, draw it and give one strength and one weakness.

# 4 Verbal and non-verbal communication

**Verbal communication** uses words. It is **oral** (conversations, phone calls, meetings, presentations, interviews, toolbox talks) or **written** (letters, memos, reports, email, minutes, specifications). **Non-verbal communication** is the sending of meaning without words: through the body, the voice, space, touch, time and appearance. Mehrabian's studies (1971) are often quoted as "7% words, 38% tone, 55% body language", but those figures apply only to messages about **feelings and attitudes**, and only when words and body disagree. Say this in the exam: it shows you understand the research instead of repeating a myth.

| Type | What it covers | Example |
|---|---|---|
| **Kinesics** | Body movement: gestures, posture, head movement, facial expression, eye contact. | Crossed arms in a design review read as resistance |
| **Oculesics** | Eye contact and gaze. | Steady eye contact signals confidence in an interview |
| **Paralanguage (vocalics)** | Tone, pitch, pace, volume, pauses, "um". | A rushed, quiet voice undermines a safety briefing |
| **Proxemics** | Use of space: intimate (0–0.45 m), personal (0.45–1.2 m), social (1.2–3.6 m), public (3.6 m+) (Hall, 1966). | Standing too close to a client feels aggressive |
| **Haptics** | Touch. | A firm handshake at an introduction |
| **Chronemics** | Use of time. | Arriving late to a site meeting signals disrespect |
| **Artefacts / appearance** | Dress, grooming, PPE, uniforms, office layout. | Correct PPE on site communicates safety culture |
| **Signs and signals** | Symbols, colours, hand signals. | Crane banksman hand signals; red = prohibition |

### The four kinesics types (examined)

1. **Gestures**: emblems (thumbs-up = OK), illustrators (tracing a shape while describing it), adaptors (pen tapping when anxious), regulators (raised hand to take a turn).
2. **Posture and head movement**: upright and open = confident; slumped = bored; nodding = agreement.
3. **Facial expressions**: the main channel for emotion; smile, frown, raised brows.

### Functions of non-verbal cues

- **Repeating**: nodding while saying "yes"
- **Contradicting**: "I'm fine" with a frown
- **Substituting**: hand signal "stop" on a noisy site
- **Complementing**: leaning forward while asking
- **Accenting**: hitting the table on a key point
- **Regulating**: pausing or gesturing to give the floor

### How appearance impacts the audience

First impressions form within seconds; appearance signals competence, respect for the audience and

4. **Eye contact**: confidence, honesty, attention; norms vary by culture.

role (PPE, uniform). Dress for the audience and occasion, keep grooming neat, avoid distracting accessories, and make dress match the message.

**EXAM SPOT** HOT

"Explain the term kinesics and write a note on the four major types of body language" [10, NUST Aug 2015]. Also "Leveraging non-verbal communication with success" and "How appearance impacts on the audience" were class presentation topics. Define, list with examples, add one line on culture.

## 5 Channels and directions of organisational communication

**Formal communication** follows official lines of authority (memos, reports, minutes, circulars). **Informal communication** (the **grapevine**) moves through social networks; it is fast and flexible, but prone to rumour and distortion. Managers should not try to kill the grapevine; they should feed it accurate information early.

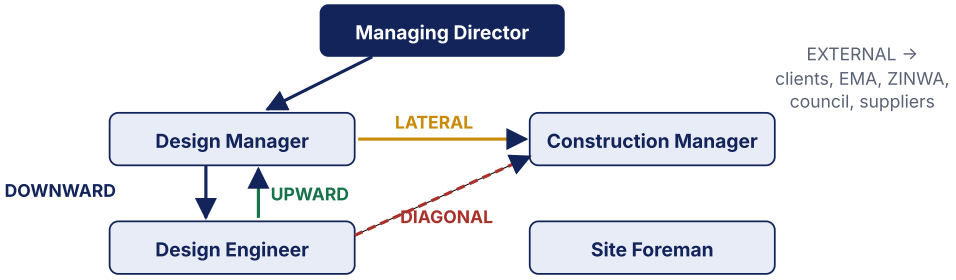



Figure 2. Directions of communication in an engineering firm.

| Direction | Purpose and examples | Problems |
|---|---|---|
| **Downward** | Superior to subordinate: instructions, policies, job descriptions, safety rules, appraisal feedback. | Distortion through many levels; overload; one-way; low morale if never two-way. |
| **Upward** | Subordinate to superior: progress and incident reports, suggestions, grievances, requests. | Fear of reprisal, filtering of bad news, managers "too busy". |
| **Lateral (horizontal)** | Same level: coordination between departments, shift handovers, design team meetings. | Rivalry, duplication, conflicting instructions, bypassing the supervisor, specialised jargon. |
| **Diagonal** | Different levels and departments with no reporting line: QA engineer to another department's foreman. | Undermines the chain of command; confusion over authority. |
| **External** | Organisation and outsiders: clients, regulators (EMA, ZINWA, local authority, SAZ), suppliers, the public. | Reputation and legal risk; must be accurate and approved. |

**Public relations (PR)** is the planned, sustained effort to build and keep goodwill and mutual understanding between an organisation and its publics (UK Chartered Institute of PR definition). Tools: press releases, media briefings, open days, corporate social responsibility, websites and social media, sponsorships, crisis communication. Functions: build reputation, manage crises, inform the public about projects (for example road closures or water cuts), support marketing and staff morale.

**EXAM SPOT** LIKELY

NUST Aug 2015 Q2: "problems associated with lateral communication" and "downward and upward flow". Dec 2014 Q2: "identify and explain communication channels in an industrial setting". "Types of communication in the workplace" and "Public relations" were class topics. For a channels question, **pick one setting (a construction site)** and use it for every example.

# 6 Barriers to communication and how to overcome them

A **barrier** is anything that blocks, distorts or delays a message between sender and receiver. Always **pair each barrier with a remedy**: examiners award marks for both halves.

| Barrier | Examples | How to overcome it |
|---|---|---|
| **Physical / environmental** | Machine noise, distance, poor lighting, bad acoustics, faulty equipment, poor network. | Brief away from running plant; use PA or radios; written backup; check equipment first. |
| **Semantic / language** | Jargon, acronyms, ambiguous words ("soon", "about"), different languages, poor grammar. | Plain language; define terms; give numbers and dates; translate (Shona/Ndebele) where needed. |
| **Psychological / emotional** | Stress, anger, fear, prejudice, low morale, poor attention, closed mind. | Pick the right moment; empathy; calm tone; separate the person from the issue. |
| **Perceptual** | Different interpretations, assumptions, selective attention, stereotypes. | Ask for feedback; check understanding; avoid assumptions. |
| **Cultural** | Different norms of eye contact, greetings, time, hierarchy, gestures. | Cultural awareness; respect; confirm meaning across groups. |
| **Organisational / structural** | Long chains of command, rigid hierarchy, unclear roles, information overload, status gaps. | Flatter structures, open-door policy, clear reporting lines, short focused messages. |
| **Physiological** | Hearing or sight problems, fatigue, illness, hunger. | Adapt channel (written plus visual), schedule wisely, use larger print or amplification. |
| **Technological** | Unfamiliar software, poor connectivity, no non-verbal cues in text. | Training; choose the simplest tool; follow important email with a call. |
| **Sender / receiver faults** | Poor planning, wrong channel, poor listening, no feedback, information overload. | Plan with the 7 C's; active listening; read-back of critical instructions. |

**MODEL PARAGRAPH · ONE BARRIER, FULLY DEVELOPED (USE THIS PATTERN FOR EVERY BARRIER)**

**Semantic barriers** arise when the sender and receiver attach different meanings to the same words or symbols. On a construction project, a structural engineer who tells the foreman to "check the cover" may mean the concrete cover to reinforcement, while the foreman thinks of the formwork cover; an acronym such as "DPC" may be unknown to a new labourer. The result is rework or an unsafe structure. This barrier is reduced by using plain language, defining technical terms the first time they are used, giving exact values ("40 mm cover to the bottom bars") and asking the receiver to repeat the instruction back.

**EXAM SPOT** HOT

Appears on almost every communication skills paper (NUST Dec 2014 Q6; UZ, MSU and CUT papers routinely). For 20 marks: definition (2), five barriers each defined, illustrated and paired with a remedy (5 × 3 = 15), conclusion (3). A class presentation "Communication barriers in communication" covered this.

# 7 The Seven C's of effective communication

The Seven C's are a checklist for any message, written or oral. The commonly taught set comes from business communication texts; an earlier list by Cutlip and Center (1952) used different C's (credibility, context, content, clarity, continuity, channels, capability of the audience), so name your source as "business communication writers".

| C | Meaning | Engineering example |
|---|---|---|
| **Clarity** | One clear purpose, simple words, one meaning only. | "Tolerance ±0.5 mm", not "fairly tight" |
| **Conciseness** | No padding or repetition; short sentences. | "Please attend" not "You are kindly requested to make yourself available to attend" |
| **Concreteness** | Specific facts and figures, not vague claims. | "Slump test failed at 140 mm on 21 Oct" not "concrete was bad" |
| **Correctness** | Accurate facts, figures, grammar, spelling, format. | Checked calculations; correct units and names |
| **Completeness** | All the facts the receiver needs: who, what, when, where, why, how. | Work order with location, quantity, deadline, contact |
| **Consideration** | Written from the receiver's view ("you-attitude"); their level and needs. | Less jargon for the client than for fellow engineers |
| **Courtesy** | Polite, respectful, tactful, unbiased. | Raising a design error without blaming the designer |

Some lists add **Coherence** (logical flow) or **Credibility**. Mention it in one line; do not replace one of the seven.

**EXAM SPOT** HOT

"Seven C's are the seven most useful qualities of effective communication. Discuss." [20 marks, NUST Aug 2015 Q5]. Intro (2), each C defined plus example (7 × 2 = 14), why they matter and a short conclusion (4). Learn the mnemonic **"5 Cs + 2 Cons"**: Clarity, Correctness, Completeness, Concreteness, Courtesy + Conciseness, Consideration.

## 8 Listening: hearing versus listening

**Hearing** is the passive, physical process of sound waves reaching the ear; it is automatic and needs no effort. **Listening** is the active, mental process of receiving, attending to, understanding, remembering and responding to spoken and non-verbal messages. You can hear without listening (background radio), but you cannot listen without hearing or seeing (sign language is "listened" to with the eyes).

| Point of difference | Hearing | Listening |
|---|---|---|
| Nature | Physiological, passive | Psychological, active |
| Effort and will | Involuntary, no effort | Voluntary, needs concentration |
| Senses | Ears only | Ears, eyes and mind (non-verbal cues too) |
| Outcome | Awareness of sound | Understanding, memory and response |
| Skill | Natural ability | Learned skill that improves with practice |

### The listening process (HURIER, Brownell)

1. **Hearing**: receiving the sounds
2. **Understanding**: grasping the literal meaning
3. **Remembering**: storing it
4. **Interpreting**: reading tone and context
5. **Evaluating**: judging facts versus opinion
6. **Responding**: feedback, verbal or non-verbal

### Types of listening

- **Informative / comprehensive**: lectures, briefings

### Five active listening skills

1. **Pay attention**: face the speaker, eye contact, put the phone away.
2. **Show that you are listening**: nod, "mm-hm", open posture.
3. **Provide feedback**: paraphrase ("What I'm hearing is..."), summarise, ask clarifying questions.
4. **Defer judgement**: don't interrupt or rehearse your reply.
5. **Respond appropriately**: honest, respectful, constructive.

- **Critical / evaluative**: judging a sales pitch or tender
- **Empathic / therapeutic**: a colleague with a problem
- **Appreciative**: music, a good speech
- **Discriminative**: telling sounds apart (a bearing failing)

### Barriers to listening

Noise; daydreaming; prejudice against the speaker; pseudo-listening (pretending); selective listening; interrupting; preparing your reply; information overload; fatigue; emotional words that trigger you.

**EXAM SPOT** HOT

Two class presentations were on this ("Distinguishing between listening and hearing" and "Five active listening skills", plus "TCW 1103 Topic 120 Active listening skills"). It is also a standard UZ, MSU and CUT question. Expect: "*Distinguish between hearing and listening and discuss five active listening skills*" [20].

## 9 Assertiveness, manipulators and put-down offenders

**Assertiveness** is the ability to express your needs, opinions and feelings honestly, directly and calmly, while respecting the rights of others. It sits between passive and aggressive behaviour.

| Style | Behaviour | Sounds like |
|---|---|---|
| **Passive** | Avoids conflict, puts others first, soft voice, little eye contact; builds resentment. | "It's fine, whatever you think." |
| **Aggressive** | Wins at others' expense; blames, shouts, interrupts, points. | "You always mess this up." |
| **Passive-aggressive** | Agrees on the surface, resists indirectly: sarcasm, sulking, deliberate delay. | "Sure, if that's what the boss wants..." |
| **Assertive** | Honest, calm, firm, respectful; steady eye contact; "I" statements. | "I need the drawings by Friday so the pour stays on schedule." |

### Assertive techniques

- **"I" statements**: "I feel... when... because... I would like..."
- **Broken record**: calmly repeat your point
- **Fogging**: agree with any truth in criticism, keep your position
- **Negative assertion**: accept a real fault without excuses
- **Negative inquiry**: "What exactly about the report was unclear?"
- **Workable compromise**: offer a solution both can accept
- **Saying no** without long apologies

### Manipulative communicators

Try to control others indirectly. Signs: guilt-tripping, flattery before a request, playing the victim, twisting your words (gaslighting), silent treatment, moving goalposts, "everyone agrees with me". **Response**: name the behaviour calmly, stick to facts, put agreements in writing, set boundaries, use the broken record.

### Put-down offenders

People who belittle others through sarcasm, insults "as jokes", public criticism or eye-rolling. **Response**: stay calm, don't retaliate, ask "What did you mean by that?", state the impact with an "I" statement, report repeated bullying through the proper channel.

## 10 Emotional intelligence and self-regulation

**Emotional intelligence (EI)** is the ability to recognise, understand and manage your own emotions and to recognise and influence the emotions of others. Goleman (1995) gives **five components**:

1. **Self-awareness**: knowing your emotions, strengths, weaknesses and triggers.
2. **Self-regulation**: controlling impulses and moods; thinking before acting.
3. **Motivation**: inner drive to achieve beyond money or status; persistence.
4. **Empathy**: understanding others' feelings and viewpoints.
5. **Social skills**: managing relationships, teamwork, persuasion, conflict handling.

**Self-regulation** (class presentation "Developing self-regulation / self-motivation") is managing your thoughts, feelings and behaviour: **Pause → Think → Choose the best response**. Components: self-awareness, emotional control (anger, stress, anxiety), impulse control, self-discipline. Developed through goal setting, time management, managing distractions, self-monitoring, positive self-talk and good decision-making. Engineering example: an engineer whose design is criticised listens, asks questions and improves the design instead of reacting angrily.

> **EXAM SPOT** LIKELY
> "Explain the components of emotional intelligence and show how they improve communication in the workplace." Use Goleman's five, each with a workplace line.

## 11 Conflict resolution and Kurt Lewin

**Conflict** is a disagreement arising when people or groups have, or believe they have, incompatible goals, needs, values or interests. Conflict is not always bad: **functional** conflict produces better ideas; **dysfunctional** conflict damages relationships and productivity.

**Causes in organisations**: scarce resources, unclear roles, poor communication, different goals between departments, personality clashes, unfair treatment, change, and differences in values or culture.

| Thomas-Kilmann mode | Assertive / cooperative | When to use |
|---|---|---|
| **Competing** | High / low (win-lose) | Emergencies, safety rules that cannot be negotiated |
| **Collaborating** | High / high (win-win) | Important issues, time available, need commitment |
| **Compromising** | Medium / medium (split the difference) | Deadlock, equal power, temporary fix |
| **Avoiding** | Low / low (lose-lose) | Trivial issues, tempers need to cool |
| **Accommodating** | Low / high (lose-win) | Issue matters more to the other side; keep harmony |

**Steps in resolving conflict**: (1) acknowledge the conflict; (2) meet privately in a neutral place; (3) let each side speak and listen actively; (4) identify the real issue and common ground; (5) generate options; (6) agree a solution and responsibilities in writing; (7) follow up.

**Kurt Lewin** (1890–1947), a founder of social psychology, gives three tools examined under "conflict prevention and management":

- **Force field analysis**: any situation is held in balance by **driving forces** (pushing for change) and **restraining forces** (resisting it). Prevent and manage conflict by listing both, then **reducing restraining forces** (fears, lack of information) rather than just pushing harder.
- **Three-step change model**: **Unfreeze** (explain why change is needed, involve people), **Change / move** (implement with communication, training and support), **Refreeze** (make the new way normal through policies and rewards). Much workplace conflict comes from badly communicated change.
- **Leadership climates** (Lewin, Lippitt and White, 1939): **autocratic** (leader decides; fast, but resentment), **democratic** (group participates; best morale and quality), **laissez-faire** (hands-off; least productive). Group climate affects how much conflict arises.

# 12 Leadership theories and transformational leadership

**Leadership** is the process of influencing people to work willingly towards shared goals. Communication is the tool of leadership: a leader who cannot communicate a vision cannot lead.

| Theory | Main idea |
|---|---|
| **Great man / trait** | Leaders are born with traits: intelligence, confidence, integrity, sociability, determination. |
| **Behavioural** | Leadership is learned behaviour. Ohio State and Michigan studies: task-oriented versus people-oriented; Blake and Mouton's managerial grid. |
| **Contingency (Fiedler)** | The best style depends on leader–member relations, task structure and position power. |
| **Situational (Hersey and Blanchard)** | Match style to follower readiness: telling, selling, participating, delegating. |
| **Transactional** | Exchange: rewards for performance, correction for mistakes; clear structures. |
| **Transformational** (Burns 1978; Bass 1985) | Inspires followers to go beyond self-interest for a shared vision. The **four I's**: **idealised influence** (role model), **inspirational motivation** (compelling vision), **intellectual stimulation** (encourages new ideas), **individualised consideration** (coaches each person). |
| **Servant** (Greenleaf) | Leader serves the team first: listening, empathy, growth of others. |

**Interpersonal communication for managers**: give clear instructions, listen, give feedback privately and specifically, praise in public, keep an open door, adapt style to each person, manage conflict early.

# 13 Whitepaper 1 exam spotter and practice questions

**ONE-PAGE SUMMARY**

- Communication = transfer + **shared understanding**; feedback makes it two-way.
- Process: sender → encoding → message/channel → decoding → receiver → feedback; noise anywhere.
- Models: Shannon-Weaver (noise, linear), Berlo SMCR (SKACS), Schramm (fields of experience), transactional (simultaneous).
- Non-verbal: kinesics (gestures, posture, face, eyes), paralanguage, proxemics, haptics, chronemics, appearance.
- Directions: downward, upward, lateral, diagonal, external; formal vs grapevine.
- Barriers: physical, semantic, psychological, perceptual, cultural, organisational, physiological, technological. **Always pair with a remedy.**
- 7 C's: clear, concise, concrete, correct, complete, considerate, courteous.
- Hearing = passive/physical; listening = active/mental (HURIER). Five active listening skills.
- Assertive = honest and respectful; "I" statements, broken record, fogging.
- EI (Goleman): self-awareness, self-regulation, motivation, empathy, social skills.
- Conflict: Thomas-Kilmann five modes; Lewin force field + unfreeze-change-refreeze.
- Transformational leadership: the four I's.

**PRACTICE QUESTIONS IN PAST-PAPER STYLE**

1. (a) Define engineering communication. [5] (b) With the aid of a well-labelled diagram, explain the elements of the communication process. [15]
2. Discuss what you understand by barriers to communication, giving examples, and examine ways of reducing their effects. [20]
3. "Seven C's are the seven most useful qualities of effective communication." Discuss. [20]
4. (a) Distinguish between hearing and listening. [6] (b) Discuss five active listening skills an engineering student can use in lectures and site meetings. [14]
5. (a) Explain the term kinesics. [4] (b) Write a note on four major types of body language. [8] (c) Explain how appearance impacts on an audience. [8]
6. Using an engineering organisation of your choice, explain downward, upward, lateral and diagonal communication and the problems associated with lateral communication. [20]
7. Compare passive, aggressive and assertive communication, and describe four assertive communication techniques. [20]
8. With reference to Kurt Lewin, discuss how conflict can be prevented and managed in organisations. [20]
9. Explain the five components of emotional intelligence and show how each improves workplace communication. [20]
10. "Transformational leadership is the most suitable style for engineering teams." Discuss. [20]

**MODEL ANSWER OUTLINE · Q4 HEARING VS LISTENING AND ACTIVE LISTENING [20]**

**(a)** Hearing is the involuntary physiological reception of sound by the ear; listening is the voluntary psychological process of attending to, understanding, remembering and responding to a message. Give three contrasts from the table (nature, effort, outcome) with an example: a student hears the generator outside but listens to the lecturer's explanation of shear force. **(b)** One paragraph per skill (about 3 marks each): pay attention, show you are listening, give feedback (paraphrase and question), defer judgement, respond appropriately. Each paragraph: what it is, how to do it, a lecture or site example, and the benefit (fewer errors, better notes, trust). Close with one sentence: listening is the half of communication most people never train.

## Sources used in this whitepaper


Barnlund, D.C. (1970) 'A transactional model of communication', in Sereno, K.K. and Mortensen, C.D. (eds) *Foundations of Communication Theory*. New York: Harper & Row.

Bass, B.M. (1985) *Leadership and Performance Beyond Expectations*. New York: Free Press.

Berlo, D.K. (1960) *The Process of Communication*. New York: Holt, Rinehart and Winston.

Brownell, J. (2012) *Listening: Attitudes, Principles and Skills*. 5th edn. Boston: Pearson.

Goleman, D. (1995) *Emotional Intelligence*. New York: Bantam.

Hall, E.T. (1966) *The Hidden Dimension*. New York: Doubleday.

Lewin, K. (1951) *Field Theory in Social Science*. New York: Harper & Row.

Mehrabian, A. (1971) *Silent Messages*. Belmont, CA: Wadsworth.

National University of Science and Technology (2014; 2015) *TIE 1102 Engineering Communication Skills*, examination papers, December 2014 and August 2015. Bulawayo: NUST.

Shannon, C.E. and Weaver, W. (1949) *The Mathematical Theory of Communication*. Urbana: University of Illinois Press.

Thomas, K.W. and Kilmann, R.H. (1974) *Thomas-Kilmann Conflict Mode Instrument*. Tuxedo, NY: Xicom.

TCW 1103 class presentations (2026): Ncube, S. *Developing self-motivation*; plus the lecturer's topic list (active listening, assertiveness, manipulators, Lewin, EI, leadership, PR, barriers).